EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Library Book Issue Request Form — Security and Governance Use Case

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1. Aim

To plan and apply security features and governance and administration practices in the Library Book Issue Request Form application built in Zoho Creator — covering sign-in security, user access, data permissions, safe scripting and integrations, controlled changes and auditing — so that member data is protected and the application is managed in an orderly way.

The application stores personal details such as the member's name, roll number, email, contact number and hostel / residential address, together with the books requested, so access control and accountability are essential.

2. Procedure

Step 1: Identify what needs protection — The library management form (book issue requests) and its reports and dashboard were reviewed, and the sensitive data was listed: member identity, roll number, email, contact number, residential address and request history.

Step 2: Plan account security — Sign-in protection was planned for all users: strong passwords, two-factor sign-in and single sign-on where the institution uses it (Section 3.1).

Step 3: Design roles and permissions — Four user types — Library admin, Student, Faculty and Staff, matching the roles created in Experiment 8 — were defined, and a permission matrix was prepared following the principle of least privilege (Section 3.2).

Step 4: Define the access flow — The checks made when a user opens the application were drawn as a decision flow (Section 3.3).

Step 5: Secure scripts and connections — Safe Deluge practices were drafted, such as validating input on the request form and keeping any external connection limited to administrators, in line with the workflow work in Experiment 10 (Section 3.4).

Step 6: Set up governance — Controlled changes through development, stage and production environments, audit review and periodic access reviews were planned (Section 3.5).

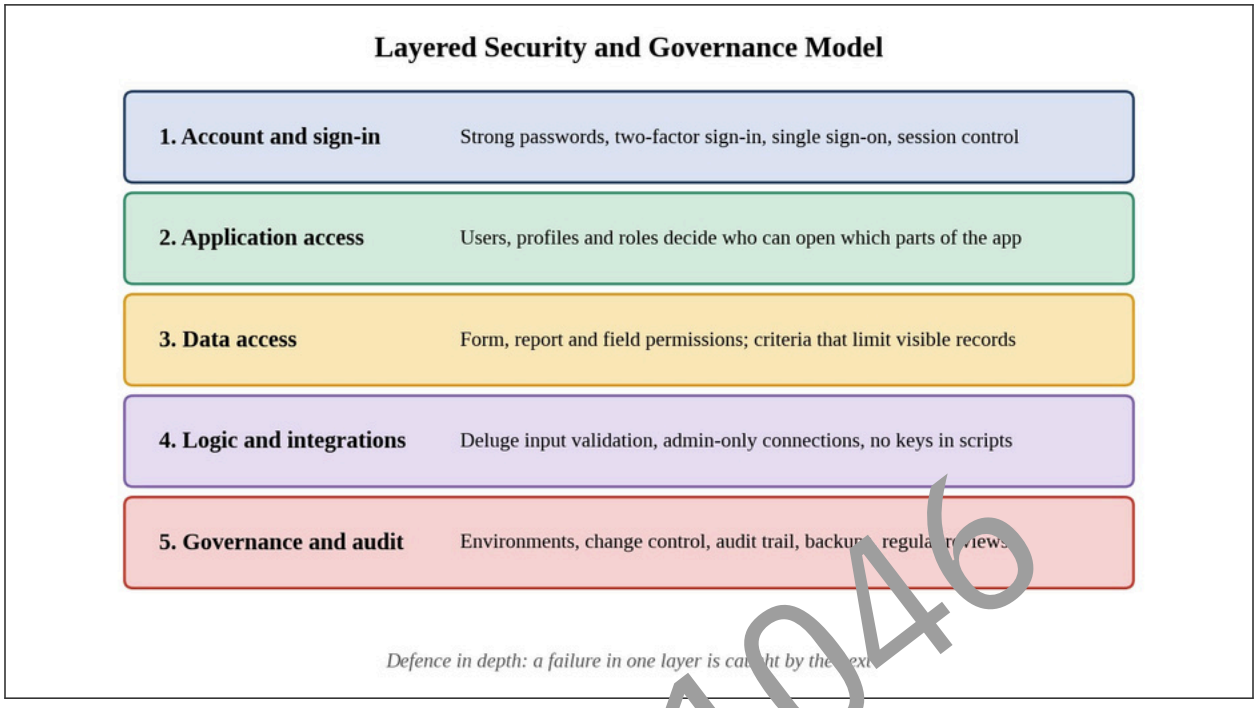
Step 7: Prepare the admin checklist — Threats and matching controls were tabulated, and an administration checklist was prepared for regular use (Sections 3.6 and 3.7).

Note: the settings are described as a planned configuration, except for the roles, permission sets and data-sharing rule that were actually set up in Experiment 8. No screenshots of the other settings were captured for this record, so it does not claim that every option was turned on. Feature names and availability depend on the Zoho Creator plan and version and should be confirmed in the current documentation.

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3. Output

3.1 Layered security and governance model



Security is arranged in five layers: account sign-in, application access, data access, logic and integrations, and governance with audit. A failure in one layer is caught by the next, which is the idea of defence in depth.

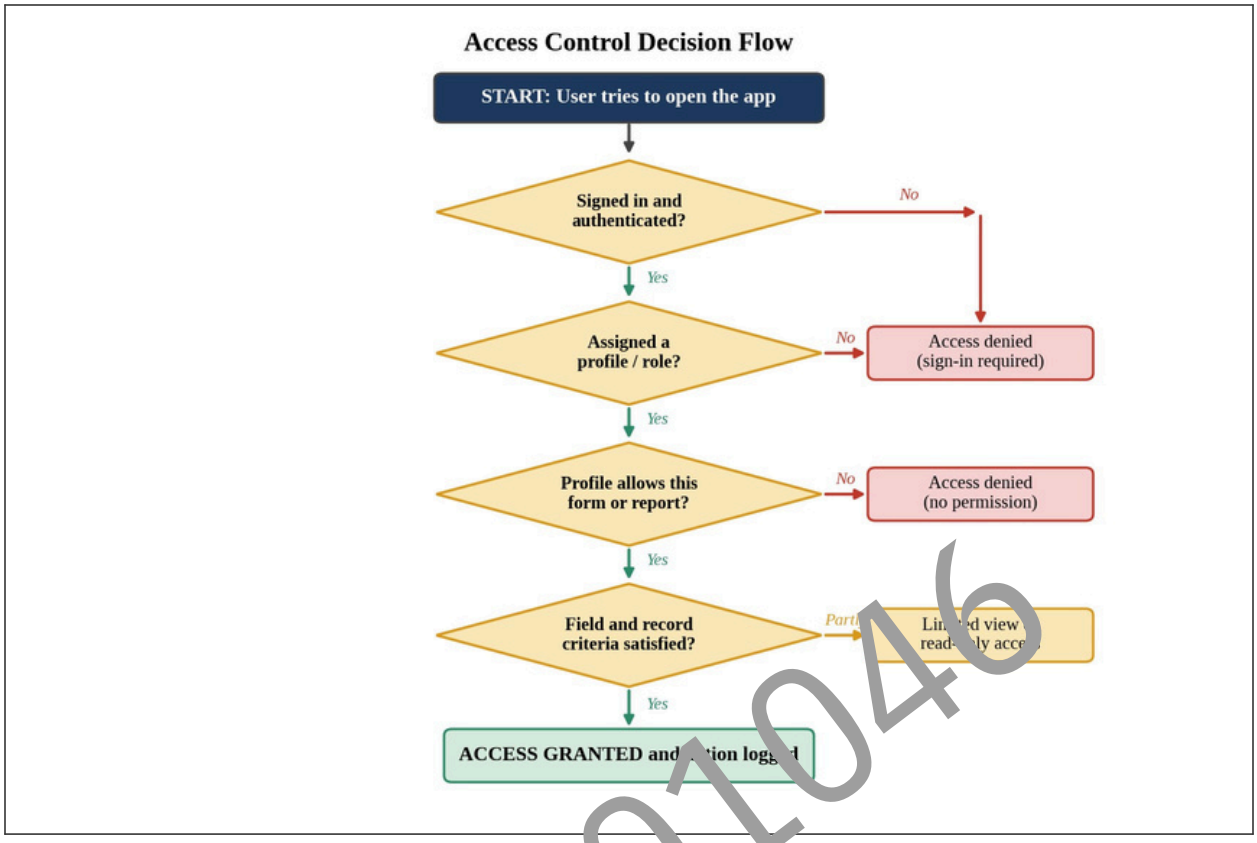
3.2 Proposed permission matrix

Each user type receives only the access needed for the job (least privilege). The matrix is a proposed design for the book issue request system; the library admin's Read/Write access to student requests follows the data-sharing rule from Experiment 8, and user roles, sharing rules and workflows are managed only by the application owner (CEO role).

Form / report	Library admin	Student	Faculty	Staff
Book issue requests (library management)	View, edit	Create and view own request	Create and view own request	Create and view own request
Member details (contact number, address)	View, edit	Own entry only	Own entry only	Own entry only
Book category and copies (request fields)	View, edit	Enter own request	Enter own request	Enter own request
Dashboard and Kanban views	View all	No access	No access	No access
Reports and exports	All reports, export	Own requests only	Own requests only	Own requests only
Users, roles, sharing rules, workflows	No access	No access	No access	No access

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3.3 Access control decision flow



When a user tries to open the application, the system checks sign-in, then the assigned profile or role, then permission on the form or report, and finally field and record criteria. Failing any check denies or limits access, and granted actions are logged.

3.4 Secure scripting and connection practices (illustrative)

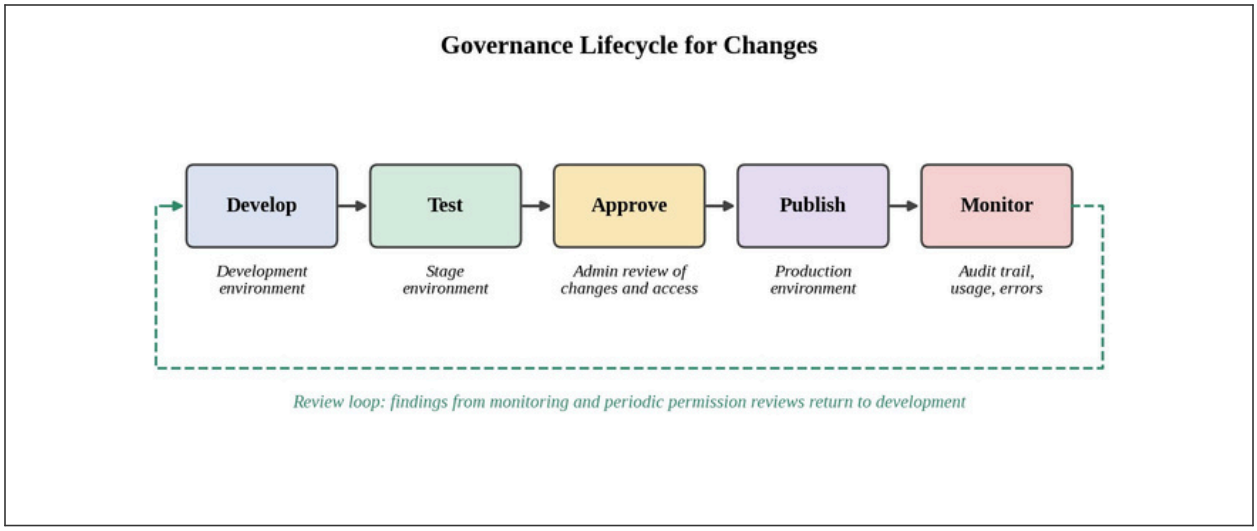
```
// On validate of the library management form
if(input.Email.is_blank())
{
    alert "Email address is required to confirm the request.";
    cancel submit;
}
if(input.Number_of_copies < 1 || input.Number_of_copies > 5)
{
    alert "Please request between 1 and 5 copies.";
    cancel submit;
}

// Connection: use the stored link name, never a key written in the script
```

Input is validated on the server side of the form so wrong or missing data is rejected. Any credentials needed for an external service stay in a connection, which is set to administrators only, and values such as the copy limit should be stored in a setting instead of being written inside the script. The field names and the limit of 5 copies are assumptions for illustration, and the code is illustrative only.

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3.5 Governance lifecycle for changes



Changes move from the development environment to the stage environment for testing, then to an administrator review, and only then to the production environment. Monitoring through the audit trail and periodic permission reviews feed improvements back into development.

3.6 Threats and matching controls

Threat	Example in this application	Control
Unauthorised access	A former student opens member contact data.	Two-factor sign-in, prompt user removal, role-based access.
Excess privilege	A student can view or export all requests.	Least privilege, separate permissions for delete and export.
Data leakage	Reports or forms shared publicly by mistake.	Review sharing settings; avoid public links.
Insecure scripts	Missing validation or keys inside Deluge code.	Validation on submit, connections instead of keys, code review.
Uncontrolled change	A live fix breaks request submission.	Dev, stage and production with approval.
No traceability	Nobody knows who changed a record.	Audit trail review and periodic log checks.

3.7 Administration checklist

Area	Administration practice	How often
Users	Add users with the right role; remove leavers the same day.	On every change
Permissions	Review roles and the sharing rule against the matrix.	Every quarter
Sign-in	Confirm two-factor sign-in or single sign-on is enforced.	Every quarter
Connections	Check connections are administrator-only and needed.	Every quarter
Changes	Publish only after testing in the stage environment.	Every release
Audit	Look through the audit trail and usage metrics for unusual activity.	Monthly
Backup and recovery	Keep a recent backup and test that data can be restored.	As planned

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4. Final Result

A layered security and governance plan was prepared for the Library Book Issue Request Form application, covering sign-in protection, role-based access with a least-privilege permission matrix, an access-control decision flow, secure scripting and connection practices, controlled changes through environments, and regular auditing.

Threats such as unauthorised access, excess privilege, data leakage, insecure scripts, uncontrolled change and lack of traceability were matched with controls, and an administration checklist was drawn up for regular use.

The exercise met its objective of designing security and governance for the application. Apart from the roles and sharing rule configured in Experiment 8, the settings were not captured as screenshots, so the record presents them as a planned configuration to be confirmed in the live application.

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